COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:		**DATE SUBMITTED:**	
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ACTIVITY NO. 1 **Configure Network using Virtual Machine**

1. PROCEDURE OUTPUT

Task 1: Changing Hostname

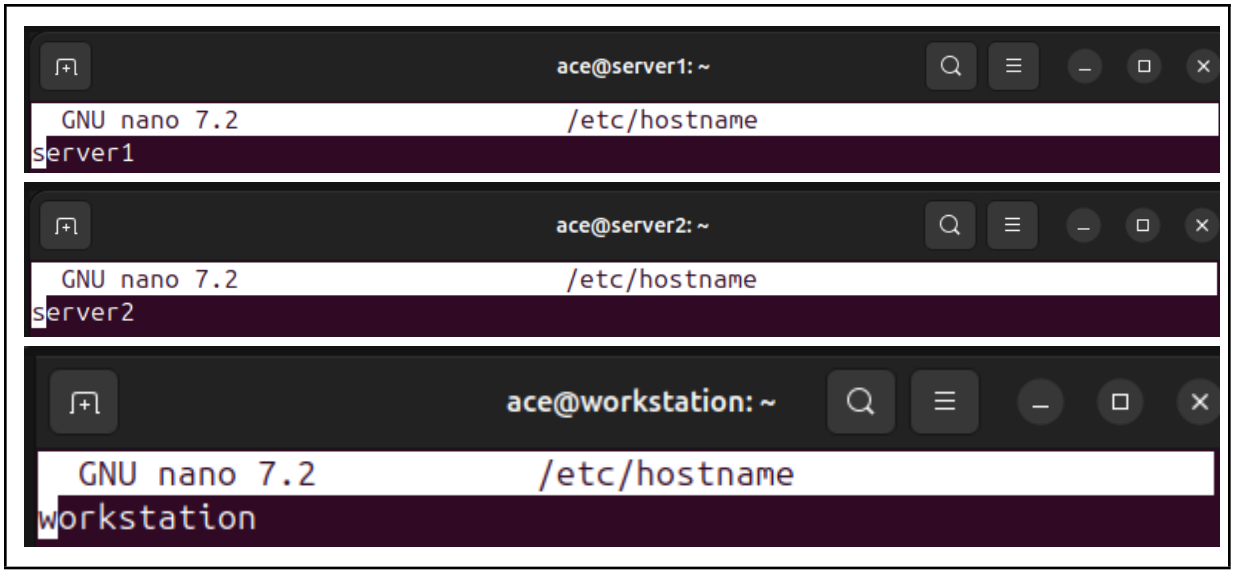
Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press Control + O to write out (save the file). Press Enter when asked for the name of the file. Press Control + X to end.

1. Change the hostname using the command sudo nano /etc/hostname

1.1 Use server1 for Server 1

1.2 Use server2 for Server 2

1.3 Use workstation for the Local Machine

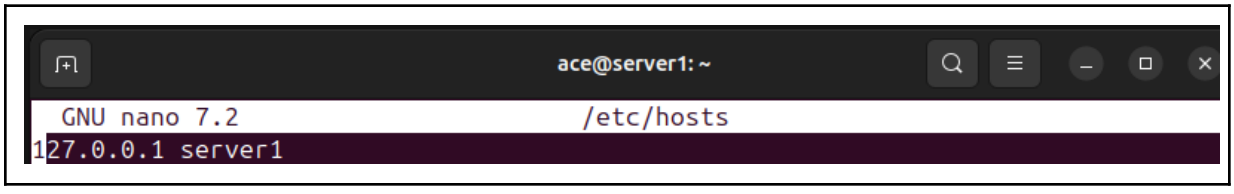


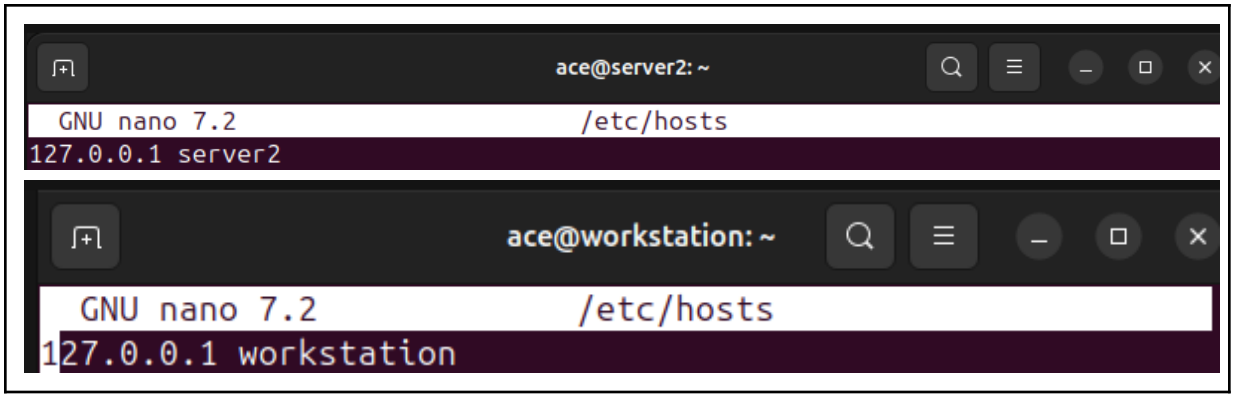
2. Edit the hosts using the command sudo nano /etc/hosts. Edit the second line.

2.1 Type 127.0.0.1 server 1 for Server 1

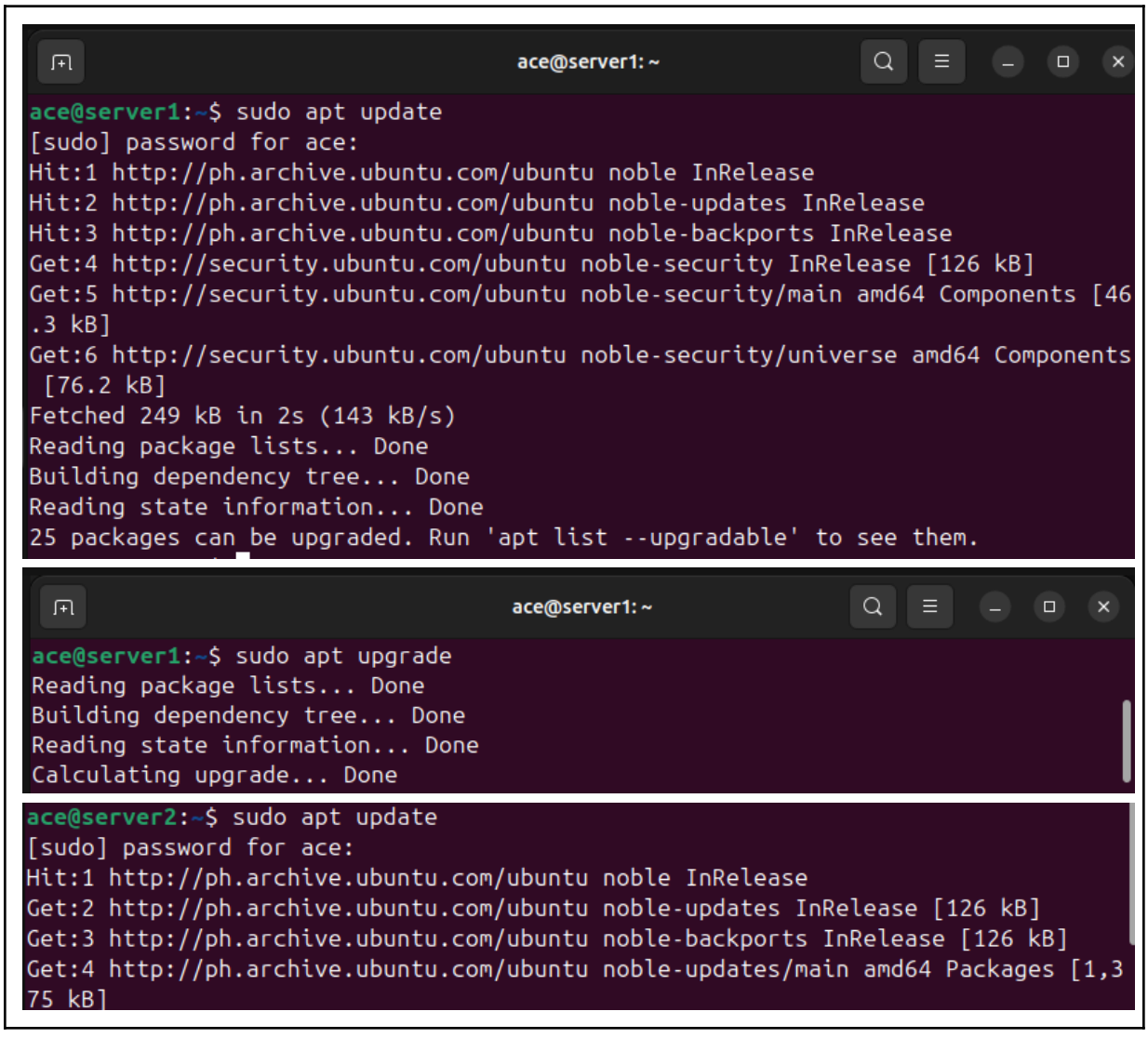
2.2 Type 127.0.0.1 server 2 for Server 2

2.3 Type 127.0.0.1 workstation for the Local Machine





Task 2: Configure SSH
Configure SSH on Server 1, Server 2, and Local Machine. Do the following:
1. Upgrade the packages by issuing the commands sudo apt update and sudo apt upgrade, respectively.



```
ace@server2:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
ace@orkstation:~$ sudo apt update
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
21 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
ace@orkstation:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libfwupd2
```

2. Install the SSH server using the command sudo apt install openssh-server.

```
ace@orkstation: ~
ace@orkstation:~$ which ssh
/usr/bin/ssh
```

3. Verify if the SSH service has started by issuing the following commands:
3.1 sudo service ssh start
3.2 sudo systemctl status ssh

```
ace@orkstation:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libfwupd2
```

```
ace@orkstation:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena>
     Active: active (running) since Tue 2026-10-13 21:52:02 PST; 2s ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 3907 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 3909 (sshd)
      Tasks: 1 (limit: 4415)
     Memory: 2.1M (peak: 2.6M)
        CPU: 28ms
     CGroup: /system.slice/ssh.service
             └─3909 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

4. Configure the firewall to all port 22 by issuing the following commands:
4.1 sudo ufw allow ssh
4.2 sudo ufw enable
4.3 sudo ufw status

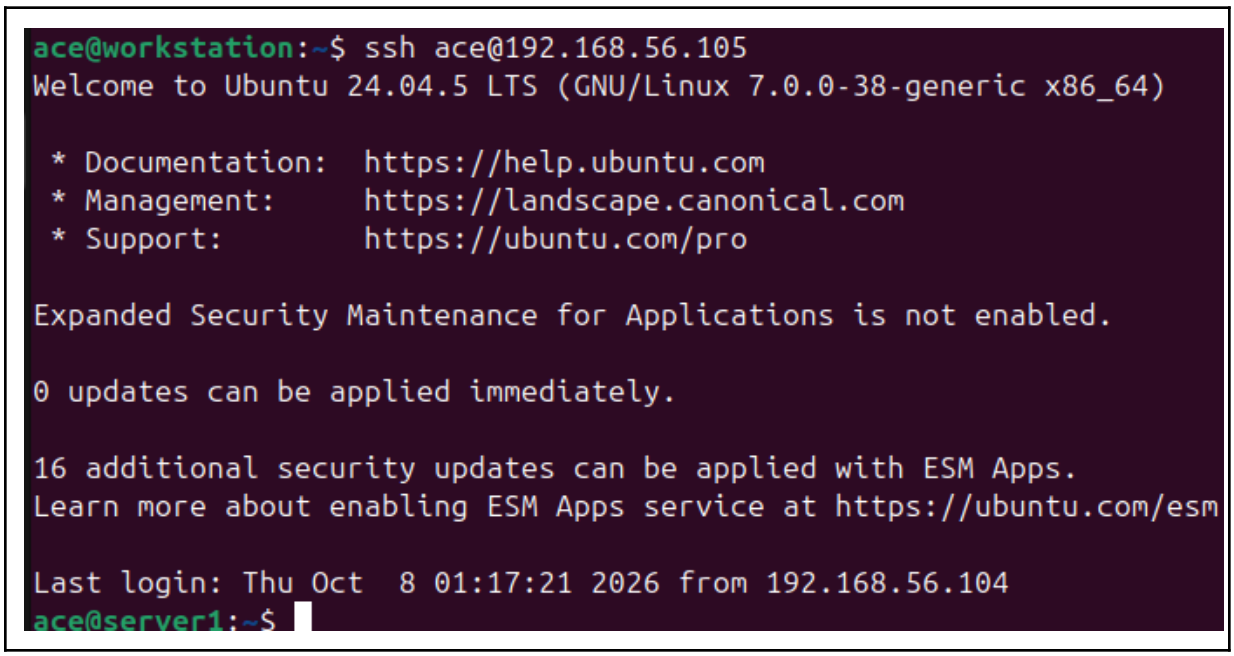
```
ace@orkstation:~$ sudo ufw allow ssh
Skipping adding existing rule
Skipping adding existing rule (v6)
ace@orkstation:~$ sudo ufw enable
Firewall is active and enabled on system startup
ace@orkstation:~$ sudo ufw status
Status: active

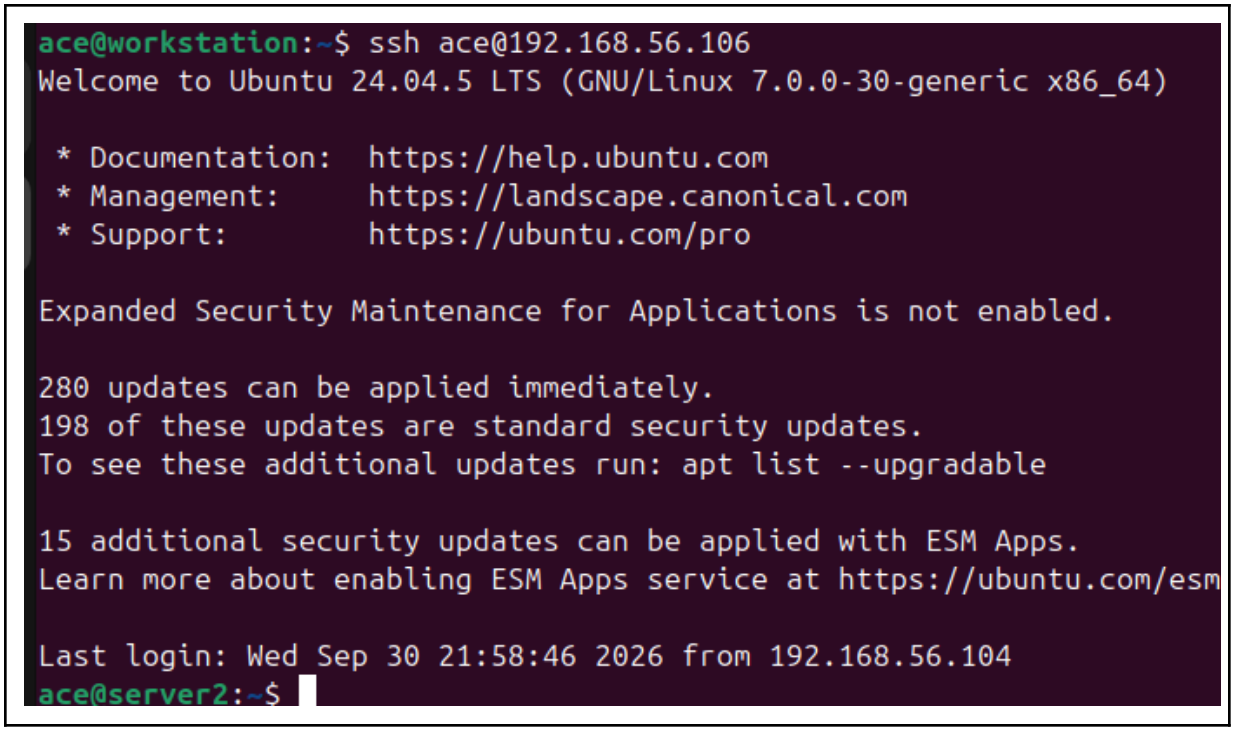
To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
80/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
80/tcp (v6)                ALLOW       Anywhere (v6)

ace@orkstation:~$
```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine.
1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.
1.1 Server 1 IP address: 192.168.56.105
1.2 Server 2 IP address: 192.168.56.106

1.3 Server 3 IP address: 192.168.56.104
2. Make sure that they can ping each other.
2.1 Connectivity test for Local Machine 1 to Server 1: ■ Successful ☐ Not Successful
2.2 Connectivity test for Local Machine 1 to Server 2: ■ Successful ☐ Not Successful
2.3 Connectivity test for Server 1 to Server 2: ■ Successful ☐ Not Successful
Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.
1. On the Local Machine, issue the following commands:
1.1 ssh username@ip_address_server1 for example, ssh jvtaylar@192.168.56.120
1.2 Enter the password for server 1 when prompted
1.3 Verify that you are in server 1. The user should be in this format user@server1.
For example, jvtaylar@server1
2. Logout of Server 1 by issuing the command control + D.
3. Do the same for Server 2.





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4. Edit the hosts of the Local Machine by issuing the command sudo nano/etc/hosts. Below all texts type the following:

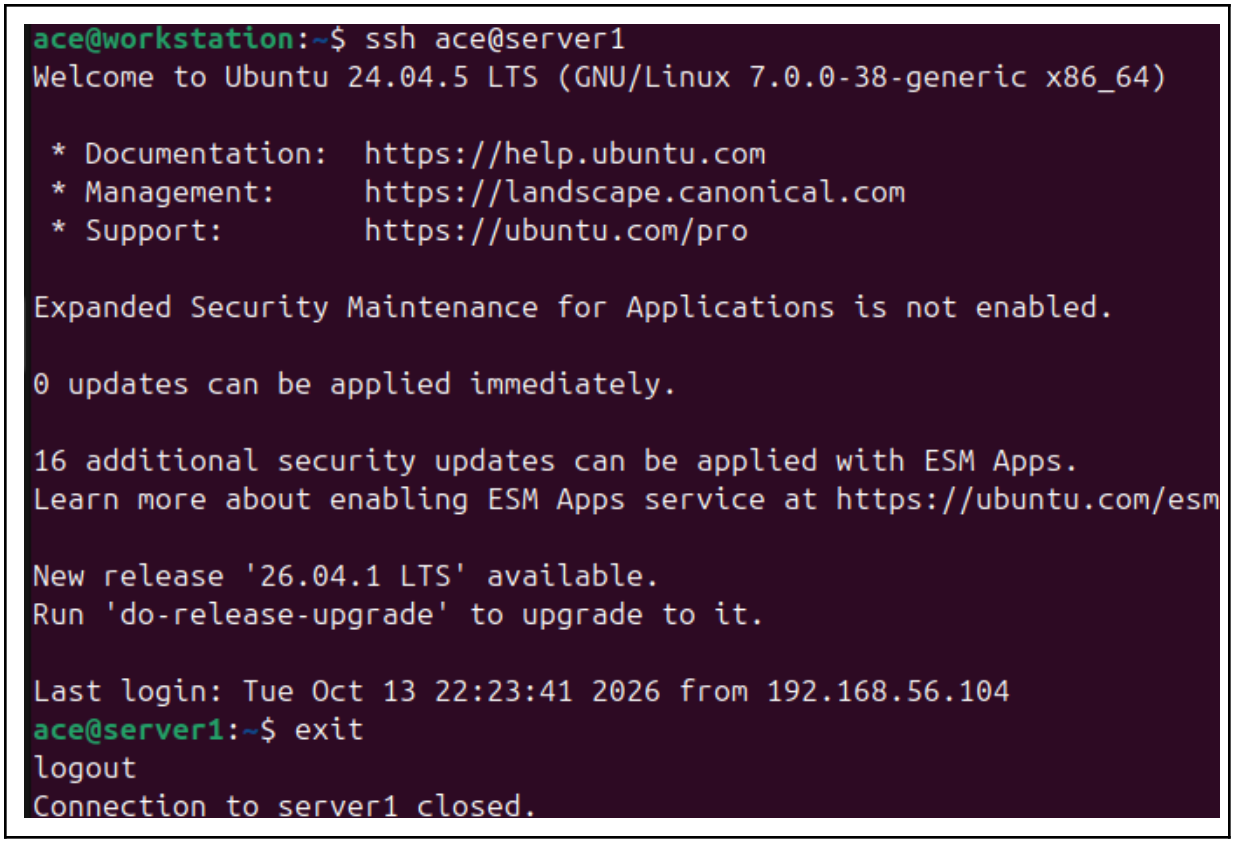
4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)

4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)

4.3 Save the file and exit.



5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do ssh jvtaylar@server1. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2


ace@workstation:~$ ssh ace@server1
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 7.0.0-38-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
16 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm
New release '26.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
Last login: Tue Oct 13 22:23:41 2026 from 192.168.56.104
ace@server1:~$ exit
logout
Connection to server1 closed.

```
ace@workstation:~$ ssh ace@server2
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

280 updates can be applied immediately.
198 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

15 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '26.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Tue Oct 13 22:24:51 2026 from 192.168.56.104
ace@server2:~$ exit
logout
Connection to server2 closed.
ace@workstation:~$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:
1. How are we able to use the hostname instead of IP address in SSH commands?
2. How secured is SSH?

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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